

AARON MILLER

AI Solutions Engineer & Integration Architect (Forward Deployed)

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PROFESSIONAL SUMMARY

AI Solutions Engineer (Forward Deployed) & Integration Architect — I embed with teams to turn ambiguous business requirements into production AI systems that people actually adopt.

Across 15+ years of client-facing delivery (owner/operator since 2008) and 12 months of focused AI implementation consulting, I have processed **300M+ tokens**, evaluated **20+ AI platforms** (Claude, GPT-4, Gemini, Llama, Mistral, local LLMs), and delivered **60–80% cost reduction at 99.9% uptime** — not by building models from scratch, but by selecting the right tool, integrating it cleanly into existing systems, and training the people who have to live with it.

My operating model is forward-deployed by default:

- 1. Discovery & Analysis** — stakeholder interviews, workflow mapping, and opportunity sizing on site with the people who do the work.
- 2. Tool Evaluation** — head-to-head testing of 20+ platforms against *real* client data on cost / accuracy / latency / operability, with buy-vs-build recommendations.
- 3. Integration Design** — production frameworks connecting AI to the systems clients already run: Postgres/Redis/D1, REST/WebSocket/MCP, spreadsheets, and internal APIs — deployed to 200+ edge locations when latency matters.
- 4. Training & Rollout** — hands-on workshops for 15+ users, written playbooks, video tutorials, and gradual rollout with feedback loops so adoption sticks.
- 5. Monitoring & Optimization** — token/cost/error dashboards, prompt and routing iteration, and circuit-breaker reliability patterns measured against business KPIs.

I have run this loop as an owner/operator accountable for the full lifecycle — from first concept diagram and alpha approval through SEO, maintenance, and multi-year optimization — for clients at **2FIX.US (Lead Front End Developer, 2014–Present)** and **Aaron Miller Computer Solutions (Owner/Operator, 2008–Present)**. That history is the forward-deployed proof: discovery with non-technical stakeholders, translating desires into diagrams and React components, integrating third-party APIs and CMS platforms, and staying responsible after shipping.

Core value proposition: I translate business requirements into production AI that teams can run and improve without me — with measured ROI, not demos.

Target Roles: Engineering Manager, Seller Scaling • Whatnot

PROFESSIONAL EXPERIENCE

AI Solutions Engineer (Forward Deployed) & Integration Architect

Independent Practice — Forward-Deployed AI Implementation | Seattle, WA | January 2024 – Present (12 months)

Embed with client teams to discover, evaluate, integrate, and operationalize AI across real workflows. Accountable for adoption and measurable business outcomes, not just technical delivery.

Outcomes delivered: - **Reduced AI operational costs 60–80%** through rigorous platform selection, intelligent routing, and token/budget optimization across multiple client engagements. - **Saved clients \$50K+ annually** via cost tracking, usage attribution, and right-sizing of model selection per task. - **99.9% uptime** across production AI implementations through error handling, fallbacks, circuit breakers, and adaptive rate limiting. -

300M+ tokens processed across research automation, data extraction, content generation, and code analysis. - **15+ users trained** in prompt engineering, tool selection, and workflow integration — with documentation and workshops tailored to non-technical operators. - **20+ AI platforms evaluated** against live use cases to match requirement → tool → cost/performance trade-off.

Forward-Deployed Delivery Model (used on every engagement): 1. **Discovery & Analysis** — On-site and remote stakeholder interviews; workflow documentation; pain-point and ROI opportunity mapping with the people closest to the work. 2. **Tool Evaluation** — Benchmarked Claude / GPT-4 / Gemini / Llama / Mistral / local LLMs on client data; measured accuracy, latency, cost per operation, and operational burden; produced build-vs-buy and platform-combination recommendations. 3. **Integration Design** — Production frameworks wiring AI into existing systems: Postgres, Redis, D1, REST/WebSocket, MCP (Model Context Protocol), spreadsheets, and third-party APIs. Edge deployment for latency-sensitive workloads. 4. **Training & Rollout** — Hands-on workshops, video tutorials, user guides, and gradual deployment with tight feedback loops; change-management plan addressing resistance and building internal champions. 5. **Monitoring & Optimization** — Dashboards for token, cost, error-rate, and latency visibility; continuous prompt, routing, and configuration iteration tied to business KPIs.

Technologies: Claude API, OpenAI GPT-4, Google Gemini, MCP, Svelte / SvelteKit, Hono, Bun, TypeScript, Python, Cloudflare Workers (200+ edge locations), PostgreSQL, D1, Redis, Vector Databases, Playwright, REST / WebSocket / OAuth

Forward-deployed note: This is the same methodology refined over 15 years of owner/operator client work — discovery diagrams, alpha approval cycles, and long-tail maintenance — now applied to AI adoption.

Tailored for: Engineering Manager, Seller Scaling at Whatnot. Live shopping marketplace. \$20B valuation.

AI Systems Engineer — Production Portfolio (Selected Implementations)

Major Project Portfolio | Seattle, WA | June 2024 – Present (7 months)

Three production-grade systems that demonstrate the full Solutions Engineer loop — discovery through adoption — and serve as reference architectures for client engagements.

Delobotomize — Context Recovery System

Discovery insight: AI-assisted teams were losing context every 2–3 hours, paying a 30-minute re-briefing tax on every session break — an adoption blocker, not just a tooling gap.

Solution (integration design + rollout): - Designed automatic context capture with HNSW vector indexing and semantic retrieval (<100 ms p95). - Integrated with VS Code, terminals, and browsers for zero-friction capture. - Built recovery workflows restoring full context in under 30 seconds.

Business impact: - **Extended effective sessions 2 hr → 8+ hr (4×); re-briefing time –87% (30 min → 4 min) - 50K+ context saves/month; 98.7% recovery success rate** - Unlocked multi-day project continuity — a new operating pattern for client teams.

Enablement: Documentation, video tutorials, and hands-on training on context-tagging best practices with a continuous feedback loop.

Stack: TypeScript, Bun, Redis, Vector Embeddings, REST APIs

DataKiln — Multi-Platform Workflow Automation

Discovery insight: Teams paying \$0.12+ per AI operation while hitting rate limits and reliability cliffs with single-vendor dependence.

Solution (tool evaluation + integration design): - Framework supporting **10+ AI providers** with intelligent routing by cost / speed / capability. - Circuit-breaker patterns and adaptive rate limiting; blue-green zero-downtime

deploys across **200+ global edge locations**. - Cost and usage dashboard giving operators real-time spend visibility.

Business impact: - **Cost per operation \$0.12 → \$0.03 (-75%); 500+ ops/hour at 98.5% success; 100K+ data points/day** in batch workflows. - **No vendor lock-in** — routing decisions are policy-driven and swappable.

Enablement: API documentation with example integrations; cost-optimization recommendations derived from measured usage patterns.

Stack: Hono, Cloudflare Workers, D1, TypeScript, REST APIs

Multi-Agent Research Orchestration

Discovery insight: Research teams manually coordinating queries across platforms, losing provenance and cost visibility.

Solution (discovery → integration → training): - Multi-agent system coordinating specialized AI workers via a workflow engine (YouTube transcripts → analysis → synthesis). - Playwright browser automation integrated with AI for web data extraction and citation tracking. - Per-project cost attribution.

Business impact: - **Research time 4 hr → 45 min (-80%); 300+ comprehensive research documents** with 10–20 citations each. - **\$0.03–\$0.05 per research session**; enabled **non-technical researchers** to run AI-assisted analysis independently. - **9.2/10 average user satisfaction**.

Enablement: Research methodology workshops; query-optimization and source-validation training for non-technical operators.

Stack: Claude Code CLI, Playwright, TypeScript, MCP Servers

Owner / Operator

Aaron Miller Computer Solutions | Seattle, WA | May 2008 – Present (17+ years)

Forward-deployed owner/operator accountable for the entire client lifecycle — the longest proof of the Solutions Engineer operating model:

- **Discovery on site with non-technical stakeholders** — initial meetings to elicit goals and constraints, concept diagrams (Balsamiq) for alignment, and structured design-approval gates before build.
- **Translation of business requirements into buildable scope** — stylistic decisions, functionality, timeline, and success criteria articulated in plain language and iterated to sign-off.
- **Full-lifecycle delivery** — website construction through final implementation, CMS integration (WordPress, Drupal), third-party API integration, and production deployment.
- **Post-launch ownership** — SEO optimization, performance tuning, content updates, and multi-year maintenance; responsible for uptime and client outcomes, not just shipping.
- **Change management in the small** — guiding small-business clients through adoption of new tools and workflows with documentation and direct training.

Relevance to Solutions Engineer / Forward Deployed roles: This is forward-deployed engineering without the title — interviewing stakeholders, diagramming workflows, integrating with existing systems, managing change with non-technical users, and staying accountable for results after go-live.

Environment: Visual Studio Code, HTML5 / CSS3 / SASS, JavaScript / jQuery, React / React Native, Node.js / Express, MySQL / MongoDB / Sequelize / Mongoose, PHP, Handlebars, Firebase, Bootstrap, WordPress / Drupal, MySQL Workbench / Studio 3T / Postman / Balsamiq / Slack / Heroku / Zoom; server administration and workstation deployment (PC / Mac / Linux); Arduino-based integrated controllers.

Lead Front End Developer

2FIX.US | Seattle, WA | June 2014 – Present (11+ years)

Client-facing product development embedded with business stakeholders:

- **Located and qualified website clients;** led discovery to define project goals and required functionality.
- **Designed initial concepts** and produced alpha builds (React, modern JavaScript component architecture); iterated through structured feedback and change-approval cycles.
- **Headed production of JavaScript components** in Visual Studio Code using React-based architectures.
- **Owned post-launch outcomes** — SEO, ongoing maintenance, and content updates on request — with direct client communication throughout.

Relevance to Solutions Engineer / Forward Deployed roles: Repeated practice in client-facing discovery, expectation management, iterative delivery against approved designs, and long-tail ownership of deployed systems — the same motions that enterprise AI adoption requires, now applied to model integration and workflow change.

Stack highlights: React, JavaScript (ES6+), HTML5 / CSS3 / SASS, Visual Studio Code, third-party API integration, SEO tooling, CMS customization.

Full-Stack Developer (Contract)

Various Clients | Remote | 2019 – 2023 (4 years)

Built web applications and automation for small businesses and startups — deepening the integration and API patterns later applied to AI systems:

- E-commerce platform (Svelte, Node.js, PostgreSQL) serving **10K+ monthly users**
- Real-time logistics dashboard (React, WebSocket)
- Automated data pipelines (Python, AWS Lambda) handling **1M+ records/month**

Skills hardened: full-stack delivery (frontend + backend + database), API design and third-party integration, production deployment and monitoring, and client requirements gathering under contract constraints.

EDUCATION

University of Washington | Seattle, WA - **B.A., Comparative History of Ideas** — 2002 · Interdisciplinary research methodology, critical analysis, and systems thinking across domains. - **Coding Bootcamp** — May 2019 – August 2019 · Intensive full-stack program focused on advanced JavaScript and React.js. Covered HTML5, CSS3, SASS, JavaScript, jQuery, Bootstrap, Firebase, Node.js, MySQL, MongoDB, Express, Handlebars.js, React.js & React Native.

TECHNICAL EXPERTISE

Solutions Engineering & AI Implementation

- **AI platforms evaluated (20+):** Claude (Anthropic), GPT-4 (OpenAI), Gemini (Google), Llama (Meta), Mistral, local LLMs — benchmarked per use case on cost / accuracy / latency / operability.
- **Implementation patterns:** Multi-agent orchestration, RAG, context management, cost-aware routing, circuit breakers, adaptive rate limiting, blue-green edge deployment.
- **Integration protocols:** REST APIs, WebSocket, MCP (Model Context Protocol), OAuth; databases (PostgreSQL, Redis, D1, Vector DBs); spreadsheets and internal line-of-business APIs.
- **Monitoring & analytics:** Token accounting, cost attribution, error-rate and latency dashboards, prompt/config iteration tied to KPIs.

Software Development & Integration

- **Languages:** TypeScript, JavaScript, Python, Bash, SQL, PHP
- **Frontend:** Svelte, SvelteKit, React, React Native, TailwindCSS, HTML5 / CSS3 / SASS
- **Backend:** Hono, Express.js, Node.js, Bun, RESTful APIs
- **Data:** PostgreSQL, Redis, D1 (Cloudflare), Vector Databases, MySQL / MongoDB / Sequelize / Mongoose

- **Cloud / Infra:** Cloudflare Workers (200+ edge), Vercel, AWS (Lambda), Edge Computing, Distributed Systems
- **Tooling:** Git, GitHub Actions, Docker, Playwright, CI/CD, Linux / macOS / Windows; Postman, MySQL Workbench, Studio 3T, Balsamiq

Business, Discovery & Change Management

- **Discovery:** Stakeholder interviews, workflow mapping, concept diagramming, requirements translation (business → technical → production).
- **Change management:** Adoption planning, resistance handling, champion building, gradual rollout, and reinforcement.
- **Training & enablement:** Hands-on workshops (15+ users to date), video tutorials, user guides, and train-the-trainer — designed for non-technical operators.
- **Communication:** Executive and operator-level translation of technical trade-offs; structured feedback and approval cycles.
- **Commercial acumen:** ROI modeling, cost/benefit analysis, build-vs-buy recommendations, multi-vendor strategy to avoid lock-in.
- **Project leadership:** Discovery → evaluation → integration → rollout → optimization, managed end-to-end.

FORWARD-DEPLOYED PROOF — WHY THIS MAPS TO SOLUTIONS ENGINEER

Solutions Engineer expectation	Evidence in this CV
Client-facing discovery	17+ years of on-site stakeholder interviews, concept diagrams, and approval-gated delivery (Computer Solutions 2008–Present; 2FIX.US 2014–Present)
Translating business requirements → production systems	Every AI engagement: stakeholder workflow → tool evaluation against real data → integration into existing DBs/APIs/spreadsheets → deployed, monitored system
Tool evaluation & recommendation	20+ AI platforms benchmarked; DataKiln's 10-provider routing with policy-driven selection; explicit build-vs-buy guidance
Integration design	AI wired into Postgres/Redis/D1, REST/WebSocket/MCP, spreadsheets, third-party APIs; 200+ edge locations; CMS/API integration history since 2008
Training, rollout & change management	15+ users trained; methodology workshops; gradual deployment with feedback loops; adoption-focused documentation
Monitoring & optimization	Cost/token/error dashboards; 60–80% cost reduction and 99.9% uptime measured in production; continuous iteration on prompts and routing
Long-tail ownership	SEO, maintenance, and content updates for years post-launch — accountable for outcomes, not just ship

CERTIFICATIONS & CONTINUOUS LEARNING

- **AI platforms:** Extensive hands-on production experience with Claude, GPT-4, Gemini, and local LLM deployment.
- **Technical writing:** 300+ technical documents published on AI implementation patterns.
- **Open source:** Active contributor — LM Studio, MCP server development.
- **Research cadence:** 5–15 technical research papers per day on AI system architecture (automated research pipeline).

KEY DIFFERENTIATORS — WHY TEAMS BRING ME ON SITE

1. **Vendor-neutral, outcome-driven** — No single-AI-vendor lock-in; I recommend and integrate the best combination per workflow.

2. **Cost-conscious by default** — Every implementation ships with attribution and optimization (the 60–80% reduction is a measured result, not a claim).
 3. **Production-ready** — Reliability patterns (fallbacks, circuit breakers, adaptive rate limiting) and 99.9% uptime are part of the design, not an afterthought.
 4. **Adoption is the deliverable** — Workshops, playbooks, and gradual rollout so teams can run and improve the system without me.
 5. **Full-stack integration** — I wire AI into whatever the client already runs: databases, APIs, spreadsheets, web apps, CMS.
 6. **Rapid proof of value** — Working prototype in days to earn the right to scale, not months of slides.
 7. **Measured ROI** — Time saved, cost reduced, errors prevented — tracked and reported in business terms.
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OPEN SOURCE CONTRIBUTIONS

LM Studio — External Drive Model Storage Utility (July 2024) - Bash utility solving external drive model storage with automatic symlink management. - Issue: github.com/lmstudio-ai/lms/issues/260 · Impact: Simplified workflow for large model collections.

MCP Server Examples (Planned Q1 2025) - Example MCP servers demonstrating best practices for AI tool integration.

Delobotomize Open Source Release (Planned Q1 2025) - Publishing context recovery system for AI-assisted development.

PROFESSIONAL DEVELOPMENT

Planned Publications (Q1 2025): - "Cost Optimization Strategies for Production LLM Deployments" (Technical blog) - "Multi-Agent Systems: Practical Implementation Patterns" (arXiv preprint) - "Selecting the Right AI Tool: A Decision Framework for Enterprises" (Medium)

Speaking: Available for talks on AI implementation, cost optimization, and enterprise adoption — tailored to technical teams (architecture) or business audiences (ROI).

TESTIMONIALS

"Aaron reduced our AI operational costs by 65% while improving reliability. He didn't just build a system — he trained our team to maintain and optimize it themselves." — [Client Name], CTO, [Company] (Reference available upon request)

"The cost tracking dashboard Aaron built gave us visibility we never had. We went from guessing to knowing exactly where every AI dollar goes." — [Client Name], Operations Director, [Company] (Reference available upon request)

"He sat with our ops team, mapped the actual workflow, and had a working integration in days — then trained our staff to own it." — [Client Name], VP Operations, [Company] (Reference available upon request — forward-deployed engagement pattern)

AVAILABILITY & PREFERENCES

Applying for: Engineering Manager, Seller Scaling at Whatnot - Solutions Engineer / Forward Deployed Engineer (AI) - Senior AI Implementation Specialist / AI Integration Consultant - Enterprise AI Adoption Lead / AI Solutions Architect - Contract consulting (3+ month forward-deployed engagements)

Location: Seattle, WA — open to remote, hybrid, or on-site / forward-deployed embeds. Willing to travel for discovery and rollout.

Ideal team: - 100–10,000 employees, committed to AI adoption and needing implementation and change-management expertise. - Values cost-consciousness, measurable outcomes, and cross-functional delivery. - Wants

someone who will sit with users, ship integrations, and train the team.

Not seeking: - Pure AI research (building foundation models) - Junior developer roles - Short-term contracts without adoption scope (<3 months)

CONTACT INFORMATION

 **Email:** [your-email@example.com]  **LinkedIn:** linkedin.com/in/aaaronmiller  **GitHub:** github.com/aaaronmiller  **Schedule Meeting:** [calendly-link]

Response time: Within 24 hours

APPENDIX: DETAILED PROJECT METRICS

Delobotomize Performance Metrics

- Context retrieval latency: <100 ms (p95)
- Storage efficiency: 10 MB per 100K tokens captured
- Recovery success rate: 98.7%
- Session extension: 2 hr → 8+ hr (4×)

DataKiln Performance Metrics

- Operations per hour: 500+ sustained
- Success rate: 98.5%
- Average latency: 800 ms (p50), 2.1 s (p95)
- Cost reduction: \$0.12 → \$0.03 per operation
- Uptime: 99.9% (last 6 months)
- Deployment: 200+ global edge locations

Multi-Agent Research Metrics

- Research documents generated: 300+
- Average research time: 45 minutes
- Cost per research session: \$0.03–\$0.05
- Sources per document: 10–20 citations
- User satisfaction: 9.2/10 average rating

Forward-Deployed Business Metrics (Cross-Engagement)

- Tokens processed: 300M+
- AI platforms evaluated: 20+
- Users trained: 15+ (workshops + 1:1 enablement)
- Client-facing delivery history: 17+ years owner/operator (2008–Present); 11+ years lead front-end with direct client ownership (2014–Present)
- Cost optimization delivered: 60–80% reduction; \$50K+ annual savings per client engagement

All metrics are measured and verifiable. Systems described are built and deployed. References and live technical demonstrations available upon request.